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Recent developments in text-to-video generation have significantly advanced the integration of natural language understanding with visual synthesis. These systems aim to convert descriptive sentences into coherent video sequences, supporting applications in educational animation and content creation.

OpenAI's model, **Sora**, presents a diffusion transformer architecture capable of generating high-definition, minute-long videos directly from textual descriptions. According to OpenAI's documentation and technical details, Sora focuses on maintaining temporal consistency and simulating realistic physical interactions, although it still faces challenges in modeling complex causal reasoning and real-world physics \cite{sora2024}.

Similarly, **Google DeepMind's Veo** model represents a multimodal generative system that produces 1080p videos of over a minute duration. As outlined in recent public documentation, Veo integrates audio with video and allows fine-grained editing. Its third version (released in 2025) includes synchronized audio capabilities, illustrating the potential of aligning language, visual scenes, and audio streams in educational video generation \cite{veo2024}.

Dream Machine by Luma AI introduces a lightweight solution capable of producing five-second videos from either text or image prompts. It emphasizes motion realism and camera perspective simulation. While praised for its smooth rendering of human gestures and cinematic movement, critiques have emerged regarding its lack of transparency in training data and model source details \cite{dreammachine2024}.

VideoPoet, developed by Google Research, treats video generation as a large language modeling task. As described in its release documentation, the model accepts text, images, or short videos as input and generates coherent video content. Its unique ability to animate static images and perform creative video continuation makes it particularly relevant to educational applications requiring visual continuity from symbolic prompts \cite{videopoet2023}.

Collectively, these models demonstrate how advances in natural language understanding, combined with vision and generative learning, can transform how visual content is created from textual inputs. These insights directly inform systems like Math2Visual, which aim to interpret mathematical prompts and produce corresponding animations in real-time.

@misc{sora2024,

title={Sora (text-to-video model)},

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